

Project Design Phase

Problem – Solution Fit Template

Date	15 February 2025
Team ID	
Project Name	Visualizing Housing Market Trends: An Analysis of Sale Prices and Features Using Tableau
Maximum Marks	2 Marks

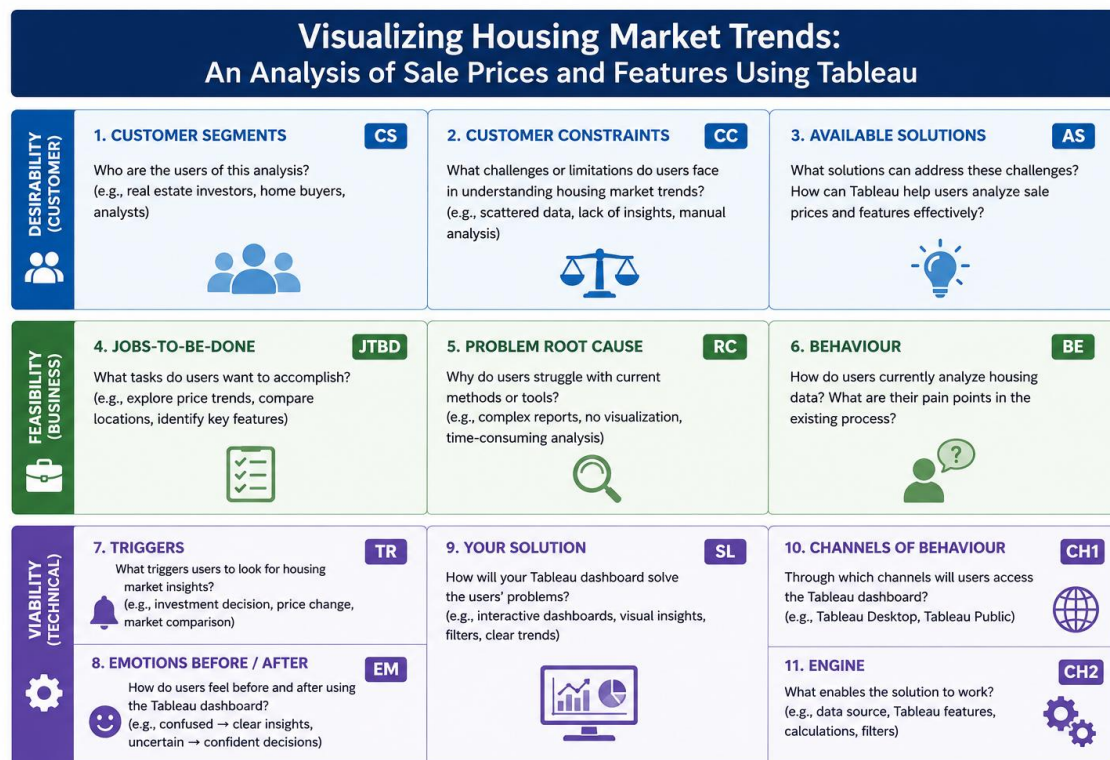
Problem – Solution Fit Template:

Problem–Solution Fit refers to identifying challenges in understanding housing market trends and developing an interactive Tableau dashboard that effectively addresses those challenges. The solution helps users analyze sale prices and housing features through clear, data-driven visualizations.

Purpose:

- ☐ Analyze housing market trends using interactive Tableau dashboards.
- ☐ Simplify complex housing data through meaningful charts and visualizations.
- ☐ Enable users to compare sale prices and property features efficiently.
- ☐ Support data-driven decision making for buyers, sellers, and analysts.
- ☐ Identify trends, patterns, and relationships between housing features and sale prices.
- ☐ Improve understanding of the housing market through easy-to-use dashboards.

Template:



References:

1. <https://www.ideahackers.network/problem-solution-fit-canvas/>
2. <https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe>

Project Design Phase
Proposed Solution Template

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Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Difficulty in understanding housing market trends and factors affecting property sale prices using raw data.
2.	Idea / Solution description	Develop interactive Tableau dashboards to analyze sale prices, property features, and market trends using visual analytics.
3.	Novelty / Uniqueness	Combines multiple housing features into interactive visual dashboards for quick and effective market analysis.
4.	Social Impact / Customer Satisfaction	Helps buyers, sellers, and real estate analysts make informed decisions based on housing market insights.
5.	Business Model (Revenue Model)	Can be used by real estate agencies, market analysts, or organizations through dashboard subscriptions and reporting services.
6.	Scalability of the Solution	Supports large housing datasets and can be extended with new regions, years, and additional property attributes.

Project Design Phase Solution Architecture

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Maximum Marks	4 Marks

Solution Architecture:

The solution architecture for this project focuses on analyzing housing market data using **Tableau**. It transforms raw housing datasets into interactive dashboards that help users understand sale price trends and property features through effective visualizations.

- Collect housing market data from CSV/Excel datasets.
- Clean and prepare the data for accurate analysis.
- Connect the processed data to Tableau for visualization.
- Create interactive dashboards to analyze sale prices, locations, and housing features.
- Generate meaningful insights to support data-driven decision making.
- Provide an easy-to-use platform for exploring housing market trends.

Example - Solution Architecture Diagram:

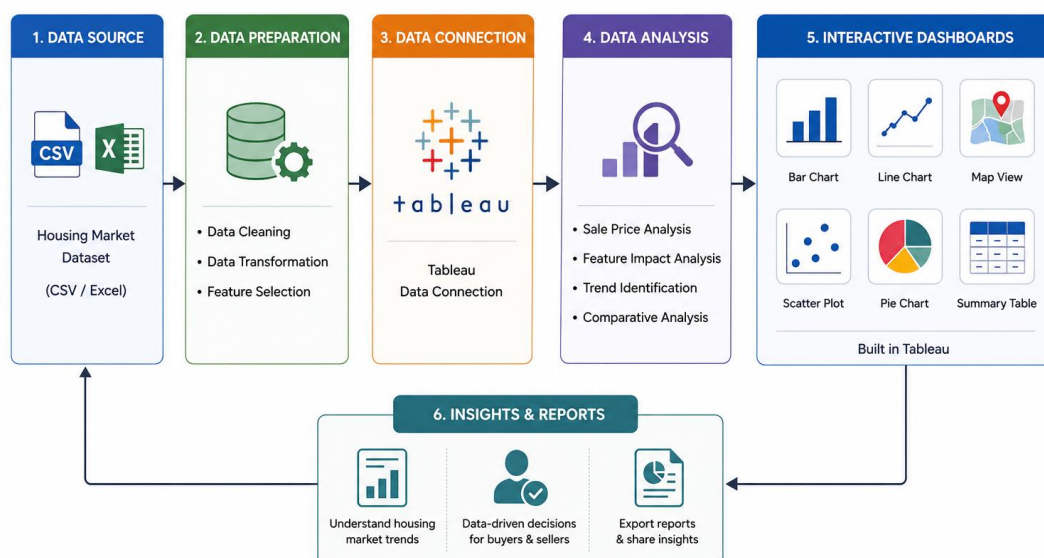


Figure 1: Architecture and data flow of the voice patient diary sample application

Reference: <https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/>